

Phase 0.1 — Trig/Log Table Interpolation

Type: Tool drill, no mission narrative. **Target time:** ~30 min if interpolation is already comfortable; longer if not. **Prerequisite:** 0.0 Slide rule basics. **Assumes:** You have a 4- or 5-place log table and a matching trig table (sine/tangent, tenths of a degree). The excerpts below are representative of real period tables in layout and precision — treat them as if torn from an actual reference; a period-correct edition (CRC Standard Mathematical Tables or similar) will be linked in `reference/tables.md` per stage as sourcing is done.

Why this matters

A table only gives you exact values at its tabulated points. Everything between two entries has to be interpolated — and every hand-computation procedure from this era (Gauss's method, Kepler's-equation solving, orbit determination worksheets) leans on this constantly. Get sloppy here and the error propagates into every later step of a multi-stage hand solution.

Method: linear interpolation

Given a table with entries at x_0 and $x_1 = x_0 + \Delta x$, and a target value x between them:

$$f(x) \approx f(x_0) + [(x - x_0) / \Delta x] \cdot [f(x_1) - f(x_0)]$$

In words: find how far across the interval your target sits (as a fraction), then move that same fraction of the way between the two tabulated values. This works well when the function is close to a straight line over the interval — which is the whole design goal of a good table (fine enough spacing that linear interpolation is trustworthy). It breaks down where the function curves sharply within one interval — see Drill 3 below.

Drill 1 — Log table interpolation

Table excerpt: $\log_{10}(x)$ for $x = 1.00$ to 1.10

x	$\log_{10}(x)$
1.00	0.00000
1.01	0.00432
1.02	0.00860
1.03	0.01284
1.04	0.01703
1.05	0.02119
1.06	0.02531
1.07	0.02938

x	$\log_{10}(x)$
1.08	0.03342
1.09	0.03743
1.10	0.04139

Interpolate:

1. $\log_{10}(1.034)$
2. $\log_{10}(1.067)$
3. $\log_{10}(1.023)$
4. $\log_{10}(1.089)$

Drill 2 — Trig table interpolation

Table excerpt: $\sin(\theta)$ for $\theta = 23.0^\circ$ to 24.0°

θ	$\sin(\theta)$
23.0°	0.39073
23.1°	0.39234
23.2°	0.39394
23.3°	0.39555
23.4°	0.39715
23.5°	0.39875
23.6°	0.40035
23.7°	0.40195
23.8°	0.40354
23.9°	0.40514
24.0°	0.40674

Interpolate:

1. $\sin(23.34^\circ)$

2. $\sin(23.78^\circ)$
3. $\sin(23.06^\circ)$

Drill 3 — When linear interpolation isn't good enough

Table excerpt: $\tan(\theta)$ near a steep region

θ	$\tan(\theta)$
80.0°	5.67128
80.5°	5.97576
81.0°	6.31375

Notice the differences aren't constant (0.30448, then 0.33799) — the function is curving noticeably even over half-degree steps. This is what “linear interpolation isn't accurate enough” looks like in practice.

1. Interpolate $\tan(80.3^\circ)$ two ways: (a) using only the 80.0° and 81.0° entries (coarse table, 1° spacing), and (b) using the 80.0° and 80.5° entries (finer table, 0.5° spacing). Record both.
2. Compare the two answers. The gap between them is the real cost of using too coarse a table near a steep region of the function — this is exactly why period trig tables near 80–90° were often tabulated at finer spacing (arcminutes, not tenths of a degree), or came with a second-difference correction column for exactly this situation.